

Mawlana Bhashani Science and Technology University

AI Native 6G Networks: Architecture, Opportunities and Challenges

Imran Hamid

Department of Information and Communication Technology

Presentation Outline

01 Introduction and research problem

02 Related work and research gap

03 Proposed solution and methodology

04 Experiments and results

05 Conclusion

Introduction and Research Problem

6G is too complex to design and run manually, so AI must be native

01 6G scale, latency and sensing demands exceed manual control

02 AI in 5G is an add-on: siloed, reactive and hard to trust

03 How can AI be native to 6G design, automation and operation?

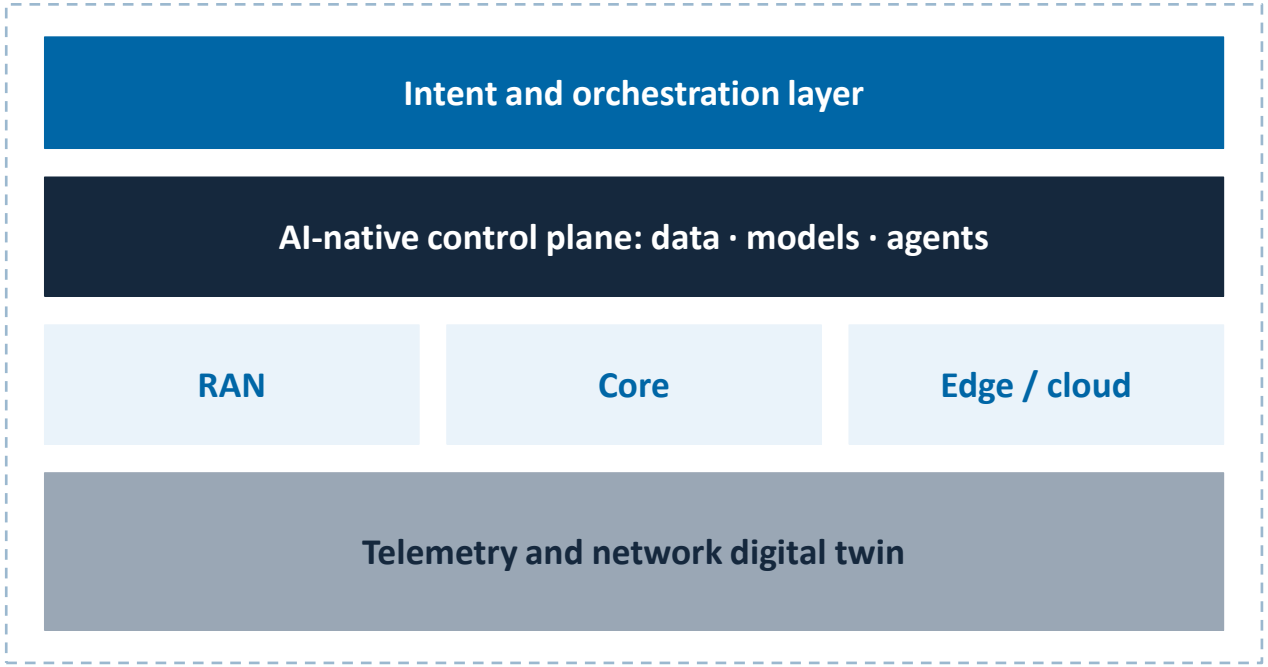
Related Work and Research Gap

Study / approach	Key idea	Relevant limitation
3GPP NWDAF [1]	Core data analytics for network functions	Core-focused and mostly reactive
O-RAN RIC xApps [2]	Pluggable AI controllers for the RAN	Siloed models; weak cross-domain control
ETSI ZSM [3]	Closed-loop zero-touch automation	Mostly rule- and policy-driven

Research gap No unified AI-native architecture spans RAN, core and edge

Methodology

AI-native 6G reference architecture



01 **Step 1: preparation**

Collect telemetry, intent and digital-twin data

02 **Step 2: core procedure**

Distributed AI agents learn, plan and coordinate

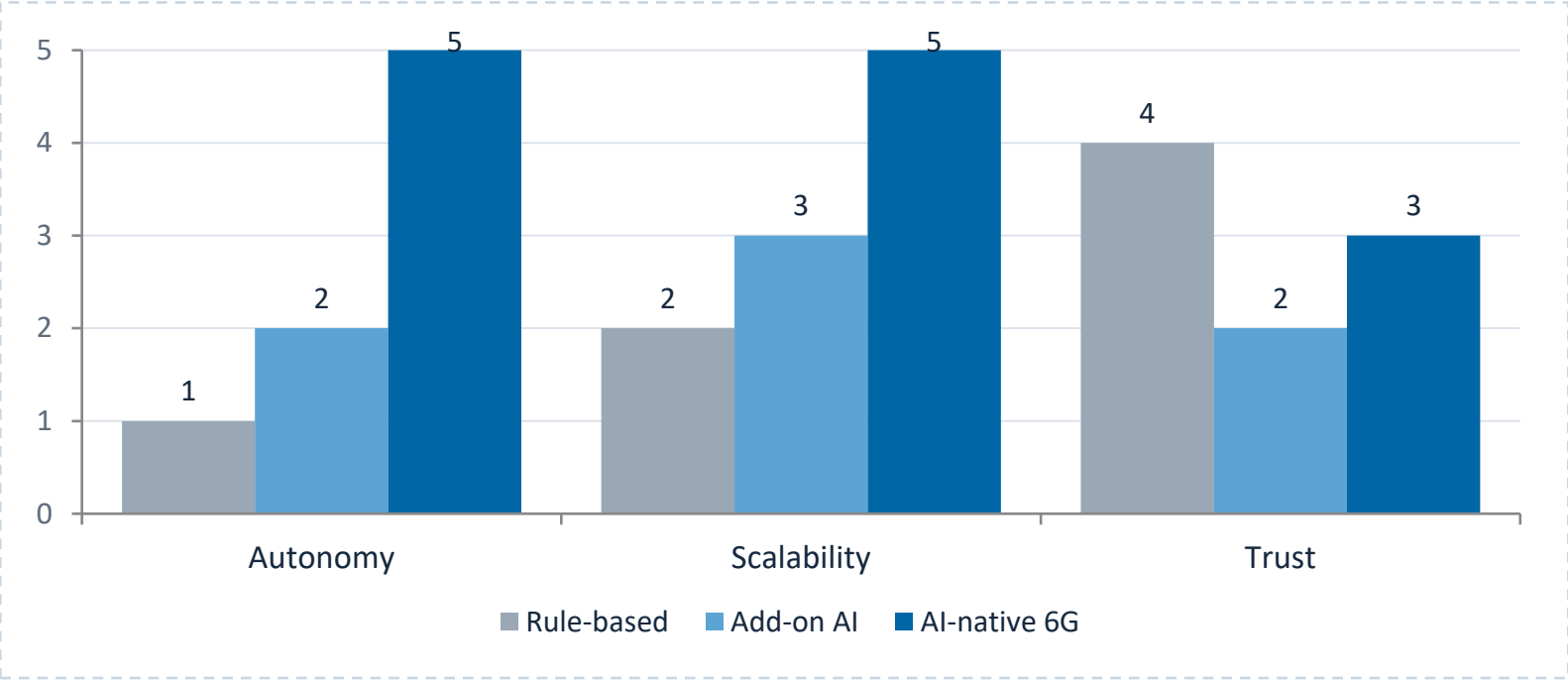
03 **Step 3: output**

Automated actions with monitoring and rollback

Experimental Setup

Data / study setting	Standards and literature: 3GPP, O-RAN, ETSI ZSM
Baselines / comparisons	Rule-based and add-on AI vs AI-native design
Evaluation metrics	Autonomy, scalability and trust (scored 1–5)
Implementation / protocol	Capability mapping across RAN, core and edge

Results and Discussion



Main finding

AI-native design leads on autonomy and scalability.

Trust and explainability remain the open challenge.

Illustrative expert scores (1–5, higher is better) for three design approaches; not measured data

Quantitative Comparison

Method	Autonomy	Scalability	Trust
Rule-based management	1 / 5	2 / 5	4 / 5
Add-on AI (5G-Advanced)	2 / 5	3 / 5	2 / 5
AI-native 6G (proposed)	**5 / 5**	**5 / 5**	**3 / 5**

AI-native leads on autonomy and scalability; trust remains open

Conclusions and Future work

AI-native design is essential for 6G, but trust and standards must mature

01 AI-native architecture enables automated 6G design and operation

02 Explainability, data privacy and standards remain open challenges

03 Next: agentic AI, digital twins and a trusted AI lifecycle

Selected References

[1] 3GPP, “Architecture enhancements for 5G System (5GS) to support network data analytics services,” TS 23.288, 2023.

[2] O-RAN Alliance, “O-RAN architecture description,” O-RAN.WG1.OAD, 2023.

[3] ETSI, “Zero-touch network and Service Management (ZSM); Reference architecture,” GS ZSM 002, 2019.

[4] W. Saad, M. Bennis and M. Chen, “A vision of 6G wireless systems: Applications, trends, technologies, and open research problems,” IEEE Network, 2020.

Thank you

Questions and Discussion

Imran Hamid
Mawlana Bhashani Science and Technology University

imraniimon14@gmail.com